

Agenda

→ Builder Design Pattern

1. A class with lot of attributes

class Student {

name;

age;

PSP;

batch;

course;

university;

grad year;

email;

password;

phone no;

⋮
⋮

}

Trade {

currency;

winTo;

coin From;

inversion Rate;

user;

base Currency;

qntity

base Exchange;

⋮
⋮

}

1. Using constructor to create object.

- Student (name, age, email);
- Student (name, age, email, password);
- Student (name, email, password);
- Student (name, email, pass, university);
- Student ("", "", "", batch);

Student (String name, String email = "0", String pass = "0");

(
Student (name);
Student (name, "an");
Student (name, "password");

default

Telescoping Constructors :

```
Student (String name) {  
    this.name = name;  
}
```

```
Student (String name, String email) {  
    this(name);  
    this.email = email;  
}
```

```
Student (name, email, pass) {  
    this(name, email);  
    this.pass = pass;  
}
```



2. Using Setters to set attributes after object creation.

```
Student s = new Student();  
s.setName("—");  
s.setAge("—");  
s.setEmail("—");  
s.validation();
```

```
class UserService {  
    login(Student s) {  
        "select * from users  
        where email =  
        s.email";  
    }  
    ↓  
    throwing Exception
```

3. Use a constructor with a single input

↓
a data structure holding
all params

Student (—)

```
{ "key" : "value"  
HashMap <String, Object> = { "name" : "Akash",  
                             "age" : 20,  
                             "email" : "akash.pr" }
```

Type Checking

```
"name" : 14  
"age" : "10"  
"email" :  
↑
```


Q. what does already exist :

- ① store diff types of params & values
- ② Comes with built in type checks

↳ Answer : Class .

```
Class StudentHelper {
```

```
    String name;
```

```
    int age;
```

```
    String email;
```

```
    String password;
```

```
    :
```

```
    all same params as Student
```

```
}
```

```
StudentHelper sh = new StudentHelper();
```

```
sh.setName("Akash");
```

```
sh.setEmail("-");
```

```
sh.setPassword("-");
```



```
Student s = new Student(sh);
```

```
Student (StudentHelper sh) {
```

```
    if (sh.email == null || sh.email == "")  
        throw Exception();
```

```
}
```

```
    this.email = sh.email;
```

```
    ;
```

```
}
```

StudentHelper → StudentBuilder

Break T:U : 8:15 AM